<u>Corollary</u> Given $v > 0$ $\exists\, m = m(n, v) > 0$, $\delta = \delta(n, v) > 0$ satisfying the following.

Suppose $(L, h) \to (M, \omega)$, $|Ric(\omega)| \leq 1$,

$\sqrt{-1}\Omega_L = \omega$ $\quad$ $Vol\, B(q, 1) > v$ $\forall q \in M$.

Then $\forall p \in M$ $\exists\, s \in H^0(M, L^m)$ s.t.

$$\|s\|_{L^2} = 1, \quad |s(p)|_{h^m} > \delta$$

<u>Proof</u> Argue by contradiction

We have $(L_k, h_k) \to (M_k, \omega_k)$

$$|Ric_{M_k}| \leq 1, \quad Vol(q, 1) > v \quad \forall q \in M_k.$$

$p_k \in M_k$. Can take GH limit

$$(M_k, p_k) \longrightarrow (Y, y, d_Y)$$

Let $\lambda_i = (\sqrt{2})^i$

up to choosing subsequence $(Y, y, \lambda_i d_Y)$ converges to tangent cone $(C(X), x^*)$

$\Rightarrow \forall \varepsilon > 0$ $\exists\, i(\varepsilon)$ s.t.

$$d_{GH}\left(B_{\lambda_i d_Y}(y, \varepsilon^{-1}), B(x^*, \varepsilon^{-1}) \right) < \varepsilon$$

Let $(\tilde{M}_k, \tilde{p}_k) = (M_k, p_k)$ scaled up by λ_i

$$\left(\tilde{L}_k = L_k^{\lambda_i^2} \text{ so } \tilde{\omega}_k = \lambda_i^2 \omega_k \right)$$

$\Rightarrow$ for $k \gg 1$

$$d_{GH}\left(B(\tilde{p}_k, \varepsilon^{-1}), B(x^*, \varepsilon^{-1}) \right) < \varepsilon, \quad |Ric_{\tilde{M}_k}| \leq \lambda_i^{-2}$$

<u>Previous Thm</u> $\Rightarrow$ if i is chosen large (i.e. ε small)

then for $k \gg 1$ $\exists\, s \in H^0(\tilde{M}_k, \tilde{L}_k^{m_1})$

$m_1 < K(n,v)$ s.t. $\|s\|_{L^2} < C(n,v)$

and $|s(\tilde{p}_k)| > \frac{1}{2}$.

Let $d = \frac{K!}{m_1}$. Then $s^d \in H^0(\tilde{M}_k, \tilde{L}_k^{K!})$

$|s| < C(n,v) \Rightarrow \|s^d\|_{L^2} \leq C(n,v)$

(by L^∞ est.) and $|s^d(\tilde{p}_k)| > \frac{1}{2^d} > \delta(n,v)$

So $s' = s^d$ is a section of $L_k^{\lambda_i^2 K!} = m$

s.t. $|s'(p_k)| > \delta(n,v)$, $\|s'\|_{L^2} < C(n,v)$ $\square$

It follows that if $L \to M$ is as in the corollary, and $\{s_0, \dots, s_N\} \subset H^0(M, L^m)$ is a basis, then

$$\Phi_{L^m} : M \longrightarrow \mathbb{CP}^N \quad \text{is well-defined}$$
$$x \mapsto (s_0(x) : \dots : s_N(x))$$

<u>Lemma</u> Let $\{s_0, \dots, s_N\} \subset H^0(M, L^m)$ be L^2-orthonormal basis.

Then $$\rho(x) = \sum_{i=0}^{N} |s_i(x)|^2 > \delta(n, v)$$

and $$N < N(m, n, v, \mathrm{Vol}(M))$$

<u>Proof</u> <u>Note</u>: $\rho : M \to \mathbb{R}$ defined above is independent of o.n. basis chosen:

if $\{\sigma_0, \dots, \sigma_N\}$ is another o.n. basis, then $\exists$ unitary matrix A_{ij} s.t. $s_i = A_{ij}\sigma_j$

$$\Rightarrow \sum_{i=0}^{N} |s_i(x)|^2 = \sum_{i=0}^{N} \underbrace{|A_{ij}\sigma_j(x)|^2}_{A_{ij}\sigma_j \overline{A_{ik}\sigma_k} h = \sigma_j \bar{\sigma}_j h = |\sigma_j(x)|^2}$$

Let $x \in M$, We can choose o.n basis $\{\sigma_i\}$ s.t. $\sigma_i(x) = 0$ for $i = 1, \dots, N$

so $\rho(x) = |\sigma_0(x)|^2$

L^∞-est. $\Rightarrow |\sigma_0| < C(n, v)$

so $\rho(x) < C(n, v)$. $\forall x \in M$.

$$1 + N = \int_M \rho \, (m\omega)^n < C(n, v)\, m^n \, \mathrm{Vol}(M, \omega)$$

shows 2nd claim.

For 1st claim, let $x \in M$.

Previous result $\Rightarrow \exists\ \sigma_0 \in H^0(M, L^m)$

s.t. $\|\sigma_0\|_{L^2} = 1$, $|\sigma_0(x)|^2 > \delta(n, v)$

extend to o.n. basis $\{\sigma_0, \sigma_1, \dots, \sigma_N\}$

$$\rho(x) \geq |\sigma_0(x)|^2 > \delta(n, v) \qquad \square$$

(Lower bound $\rho > \delta(n, v)$ is called Tian's partial C^0-estimate)

<u>Lemma</u> $\Phi_{L^m} : (M, m\omega) \to (\mathbb{CP}^N, \omega_{FS})$

defined using o.n. basis has uniform Lipschitz bound $|\nabla \Phi_{L^m}| < C(n, v)$

<u>Pf</u> Let $x \in M$. Choose o.n. basis $\{s_i\}$

s.t. $s_i(x) = 0$ for $i = 1, \dots, N$

i.e. $\Phi_{L^m}(x) = (1 : 0 : \dots : 0) \in \mathbb{CP}^N$

In chart $U_0 \subset \mathbb{CP}^N$ using coords $z^j = \dfrac{Z^j}{Z^0}$, $j \geq 1$

Then $\Phi_{L^m}(y) = s_0(y)^{-1}\left(s_1(y), \dots, s_N(y)\right)$

and at origin in U_0, $\omega_{FS} = \sqrt{-1}\,\partial\bar\partial |z|^2$

So $\nabla \Phi_{L^m}(x) = s_0(x)^{-1}\left(\nabla s_1(x), \dots, \nabla s_N(x)\right)$

$|\nabla \Phi_{L^m}(x)| < C$ using lower bound for $|s_0(x)|^2 = \rho(x) > \delta$

and gradient est. $|\nabla s_i(x)| < C$.

$\square$

Suppose $(M_k, \omega_k) \longrightarrow (Y, d_Y)$ in GH-sense

$L_k \to M_k$ has curv. $-\sqrt{-1}\,\omega_k$.

Replace L_k by L_k^m so we can assume

$\Phi_{L_k} : M_k \longrightarrow \mathbb{CP}^N$ defined for $k \gg 1$

On Y_{reg} (regular set $R \subset Y$) we have line bundle L_Y

Define $H^0(Y, L_Y^j) = \left\{ \begin{array}{l} \text{holo. sections } s \text{ of} \\ L_Y^j \text{ over } Y_{reg} \text{ with} \\ \|s\|_{L^2} < \infty \end{array} \right\}$

Cutoff fn. & L^2-estimate

$\Longrightarrow$ for $s \in H^0(Y, L_Y^j)$ $\exists\, s_k \in H^0(M_k, L_k^j)$

s.t. $s_k \to s$ under GH-convergence.

L^∞-est. on $M_k \Rightarrow |s| \in L^\infty(Y_{reg})$

In fact $\forall j$ we have $\dim H^0(Y, L_Y^j) = \dim H^0(M_k, L_k^j)$

if $k \gg 1$.